

# COMPUTATIONS ASSOCIATED WITH THE PAPER THURSTON'S SPINE IN GENUS 2

This note presents the main computational components of paper [1], including the underlying mathematical principles and the developed algorithms. The implementation of these algorithms is provided in both Python and SageMath. For detailed instructions on how to use the code, please refer to the GitHub repository: <https://github.com/a-n-n-i/compuations>.

## 1. COMPUTING ALL CURVE LENGTHS IN $C$ FROM 6 PARAMETERS

A filling subset of  $C$  must contain at least four curves. Cutting the surface along a 4-chain yields a 12-gon. According to [2], parameterizing the surface via this polygon requires three angle parameters and six edge length parameters. Given any six parameters, the remaining three can be determined numerically by the following equation:

$$(1) \quad M_{12} \dots M_1 = I \text{ in } PSL(2, \mathbb{R})$$

where  $M_i$  are matrices determined by the length parameters and angle parameters.

Numerical root-finding requires initial guesses within a reasonable error bound. Specifically, for the algorithm in this paper, we cut the surface along  $c_1, c_2, c_8, c_9$ , resulting in a 12-gon  $\mathcal{F}$  with 6 length parameters and 3 angle parameters. The algorithm for computing the lengths of all curves in  $C$  using six of these parameters is as follows:

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### Algorithm 1 Computing All Curve Lengths in $C$ from 6 Parameters

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- 1: **Input:** 6 parameters, initial values
  - 2: **Solve:** another 3 parameters of  $\mathcal{F}$  by the equation 1  $\triangleright$  Then we know all parameters of the 12-gon and the length of  $c_1, c_2, c_8, c_9$
  - 3: **Compute**  $\text{length}(c_7)$ :
  - 4:  $\mathcal{C} := \{\text{all geodesics connecting } l_1 \text{ and } l_2 \text{ that form closed geodesics}\} \triangleright$  where  $l_1$  and  $l_2$  are the pair of edges of the 12-gon corresponding to  $c_8$ .
  - 5: **Solve:**  $d\text{length}(l) = 0$  where  $l \in \mathcal{C} \triangleright$  the length of this minimal point is  $\text{length}(c_7)$
  - 6: **Compute**  $\text{length}(c_{11}), \text{length}(c_6) \triangleright$  similar with the above
  - 7: **Cut  $S_2$  along  $c_1, c_2, c_6, c_7$ :** we can get new 6 length parameters and 3 angle parameters, use these parameters, solve  $\text{length}(c_9)$
  - 8: **Repeat the above:** solve all curve lengths in  $C$
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## 2. DETERMINING THE STRATA ADJACENT TO CRITICAL POINTS

If  $C' \subseteq C$  determines a stratum  $Sys(C')$  such that the point Bolza surface is adjacent to  $Sys(C')$ , then within a small neighborhood of point  $p$  there exists a curve  $\gamma(t) = (\gamma_1(t), \gamma_2(t), \gamma_3(t), \gamma_4(t), \gamma_5(t), \gamma_6(t))$ , where  $t \in [0, \epsilon)$ ,  $\gamma(0) = p$ ,  $\gamma((0, \epsilon)) \subseteq Sys(C')$ , such that for any  $c_i, c_j \in C'$ , we have:

$$L(c_i)(\gamma(t)) = L(c_j)(\gamma(t))$$

Let  $h_i(t) = L(c_i)(\gamma(t))$ , then

$$\begin{aligned}
 h_i(t) &= L(c_i)(p) + \sum_{i=1}^6 \frac{\partial L(c_i)}{\partial x_i}(p) \gamma'(0) t \\
 &+ \frac{1}{2} \left( \sum_{i,j=1}^6 \frac{\partial^2 L(c_i)}{\partial x_i \partial x_j}(p) \gamma'_i(0) \gamma'_j(0) + \sum_{i=1}^6 \frac{\partial L(c_i)}{\partial x_i}(p) \gamma'(0) \right) t^2 \\
 &+ \frac{1}{6} \left( \sum_{i,j,k=1}^6 \frac{\partial^3 L(c_i)}{\partial x_i \partial x_j \partial x_k}(p) \gamma'_i(0) \gamma'_j(0) \gamma'_k(0) + 3 \sum_{i,j=1}^6 \frac{\partial^2 L(c_i)}{\partial x_i \partial x_j}(p) \gamma'_i(0) \gamma''_j(0) \right. \\
 &\left. + \sum_{i=1}^6 \frac{\partial L(c_i)}{\partial x_i}(p) \gamma'(0) \right) t^3 + O(t^4).
 \end{aligned} \tag{2}$$

Let  $v_1 = \gamma'(0), v_2 = \gamma''(0), v_3 = \gamma'''(0)$ , and let  $v_i = (v_i^1, \dots, v_i^6)$ ,

Therefore, a necessary condition for  $C'$  to determine a stratum adjacent to  $p$  is that for  $c_i, c_j \in C'$ , there exist vectors  $v_1, v_2$  and  $v_3$  such that:

$$\begin{aligned}
 v_1 \cdot (dL(c_i)|_p)^T &= v_1 \cdot (dL(c_j)|_p)^T \\
 v_2 \cdot (dL(c_i)|_p)^T + v_1 \cdot H_i(0) \cdot (v_1)^T &= v_2 \cdot (dL(c_j)|_p)^T + v_1 \cdot H_j(0) \cdot (v_1)^T \\
 \sum_{r,s,k=1}^6 \frac{\partial^3 L(c_i)}{\partial x_r \partial x_s \partial x_k}(p) v_1^r v_1^s v_1^k + 3v_1 \cdot H_i(0) \cdot (v_2)^T + v_3 \cdot (dL(c_i)|_p)^T \\
 &= \sum_{r,s,k=1}^6 \frac{\partial^3 L(c_j)}{\partial x_r \partial x_s \partial x_k}(p) v_1^r v_1^s v_1^k + 3v_1 \cdot H_j(0) \cdot (v_2)^T + v_3 \cdot (dL(c_j)|_p)^T
 \end{aligned} \tag{3}$$

and for small enough  $t$ :

$$\begin{aligned}
 h'_i(0)t + h''_i(0)t^2 + h'''_i(0)t^3 &= h'_j(0)t + h''_j(0)t^2 + h'''_j(0)t^3 \text{ for } c_i, c_j \in C' \\
 h'_i(0)t + h''_i(0)t^2 + h'''_i(0)t^3 &> h'_k(0)t + h''_k(0)t^2 + h'''_k(0)t^3 \text{ for } c_i \in C', c_k \in C \setminus C'.
 \end{aligned} \tag{4}$$

For each  $i$ ,  $dL(c_i)|_{M_{12}}$ ,  $H_i(0)$  and  $\frac{\partial^3 L(c_i)}{\partial x_r \partial x_s \partial x_k}$  are numerically computed<sup>1</sup>.

### 3. COMPUTING STRATUM LOCALLY AROUND $M_{12}$

The differential matrix  $D = (dL(c_1), \dots, dL(c_{12}))$  of Bolza surface can be numerically computed, it can be identified as:

$$\begin{bmatrix}
 0 & 0 & -\sqrt{2} & \sqrt{2} & -2 & 2+\sqrt{2} & 2+\sqrt{2} & -2-2\sqrt{2} & 2 & 0 & -2-\sqrt{2} & \sqrt{2} \\
 0 & 1 & -1 & 0 & -\sqrt{2}/2 & 0 & 1+\sqrt{2}/2 & -1 & 0 & \sqrt{2}/2 & -1-\sqrt{2}/2 & 1 \\
 2 & 0 & -2-\sqrt{2} & \sqrt{2} & -2-\sqrt{2} & 2+\sqrt{2} & 2+2\sqrt{2} & -2-2\sqrt{2} & 0 & \sqrt{2} & -2-2\sqrt{2} & 2+\sqrt{2} \\
 0 & 0 & 0 & 0 & 2\sqrt{1+\sqrt{2}} & -2\sqrt{2+2\sqrt{2}} & -2\sqrt{1+\sqrt{2}} & 2\sqrt{2+2\sqrt{2}} & 0 & -2\sqrt{1+\sqrt{2}} & 2\sqrt{7+5\sqrt{2}} & -2\sqrt{2+2\sqrt{2}} \\
 0 & 0 & -2\sqrt{1+\sqrt{2}} & 2\sqrt{1+\sqrt{2}} & 0 & -2\sqrt{1+\sqrt{2}} & 2\sqrt{1+\sqrt{2}} & 0 & 0 & 0 & 2\sqrt{1+\sqrt{2}} & -2\sqrt{1+\sqrt{2}} \\
 0 & 0 & 0 & 0 & 0 & -2\sqrt{2+2\sqrt{2}} & 0 & 2\sqrt{2+2\sqrt{2}} & 0 & 0 & 2\sqrt{2+2\sqrt{2}} & -2\sqrt{2+2\sqrt{2}}
 \end{bmatrix}$$

<sup>1</sup>These data are provided in the supplementary material

Let  $A = (dL(c_1), dL(c_2), dL(c_7), dL(c_8), dL(c_9), dL(c_{11}))$ , then  $\tilde{D} = A^{-1}D$  is the following:

$$\begin{bmatrix} 1 & 0 & -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & -\frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & 0 & 0 & 0 & -\frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & 1 & 0 & 0 & \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & 0 & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & 0 & 1 & 0 & \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & 0 & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & 0 & 0 & 1 & -\frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & 0 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & 0 & 0 & 0 & -\frac{1}{2} & 1 & -\frac{1}{2} \end{bmatrix},$$

which is simpler.

Up to isomorphism of  $\text{Aut}^\pm(M_{12})$ , there are three 4-chains that are adjacent to  $M_{12}$ , they are  $\{c_1, c_4, c_{10}, c_{11}\}$ ,  $\{c_1, c_4, c_{10}, c_{12}\}$ ,  $\{c_1, c_{10}, c_{11}, c_{12}\}$ , next discuss these three cases.

Let  $C' \subseteq C$ ,  $C'$  can determine a cell in "Voronoi decomposition", which can be defined as:

$$\mathcal{V}(C', M_{12}) = \{v \in T_{M_{12}}\mathcal{T}_2 : vL(c_i) = vL(c_j) < 0 \text{ for } c_i, c_j \in C', vL(c) > vL(c') \text{ for } c \in C \setminus C', c' \in C'\}.$$

Its closure is:

$$\overline{\mathcal{V}(C', M_{12})} = \{v \in T_{M_{12}}\mathcal{T}_2 : vL(c_i) = vL(c_j) \leq 0 \text{ for } c_i, c_j \in C', vL(c) \geq vL(c') \text{ for } c \in C \setminus C', c' \in C'\}.$$

Next compute the strata of the three 4-chains locally around  $M_{12}$ .

### 3.1. $\text{Sys}(\{c_1, c_4, c_{10}, c_{11}\})$ .

$$\mathcal{V}(\{c_1, c_4, c_{10}, c_{11}\}, M_{12}) = \emptyset$$

$$\begin{aligned} \overline{\mathcal{V}(\{c_1, c_4, c_{10}, c_{11}\}, M_{12})} &= \overline{\mathcal{V}(\{c_1, c_4, c_8, c_{10}, c_{11}\}, M_{12})} \\ &= \{k(-1, x, y, -1, 3 - x + y, -1)A^{-1} : k > 0, -1 \leq x \leq 3, y \geq -1, x - 4 \leq y \leq x\} \end{aligned}$$

We need to use second order derivative to get more information about this cell. For  $C_1 \subseteq C_2$ , define

$$\begin{aligned} \mathcal{V}_2(C_1, C_2, M_{12}) &= \{v_1, v_2 : v_1 = \gamma'(0), v_2 = \gamma''(0), \text{ where } \gamma(t) \text{ is a curve in } \mathcal{T}_2, \\ \gamma(0) &= M_{12}, h_i(t) = L(c_i)(\gamma(t)), \text{ such that } h'_i(0) = h'_j(0) \text{ for } c_i, c_j \in C_1, \\ h''_i(0) &= h''_j(0) \text{ for } c_i, c_j \in C_1, h''_i(0) \leq h''_j(0) \text{ for } c_i \in C_1, c_j \in C_2 \setminus C_1.\} \end{aligned}$$

To compute  $\mathcal{V}_2(\{c_1, c_{10}, c_{11}, c_{12}\}, \{c_1, c_8, c_{10}, c_{11}, c_{12}\}, M_{12})$ , let  $v_1 = k(-1, x, y, -1, 3 - x + y, -1)A^{-1}$ , where  $k > 0, -1 \leq x \leq 3, -1 \leq y \leq 3$ . Suppose  $v_2 = (\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5, \lambda_6)$ , let  $f_i(x, k) = v_1 \cdot H_i(0) \cdot (v_1)^T$ , to compute  $v_2$ , we need to solve the following linear system:

$$\begin{aligned} v_2 \cdot dL(c_i)|_p + f_i(x, k) &= a, \text{ for } i = 1, 10, 11, 12, \\ v_2 \cdot dL(c_i)|_p + f_i(x, k) &\geq a, \text{ for } i = 8, \end{aligned}$$

where  $a \in \mathbb{R}$ .

By calculation, we can obtain:

$$-f_1 - f_{10} + f_4 - f_{11} + f_8 \geq 0$$

Compute it out, we have

$$(y + 1)(y - 2x + 3) \geq 0$$

So  $y = -1$  or  $y \geq 2x - 3$ .

Thus,

$$v_1 \in \{k(-1, x, y, -1, 3 - x + y, -1)A^{-1} : k > 0, -1 \leq x \leq 3, y \leq x, y \geq -1 \text{ or } 2x - 3 \leq y\}$$

Which has these faces:

$$\begin{aligned} &\{k(-1, x, x, -1, 3, -1)A^{-1} : k > 0, -1 \leq x \leq 3\} \\ &\{k(-1, x, 2x - 3, -1, x, -1)A^{-1} : k > 0, 1 \leq x \leq 3\} \\ &\{k(-1, x, -1, -1, 2 - x, -1)A^{-1} : k > 0, -1 \leq x \leq 3\} \end{aligned}$$

So locally,  $\text{Sys}(\{c_1, c_4, c_{10}, c_{11}\})$  should look like Figure 1.

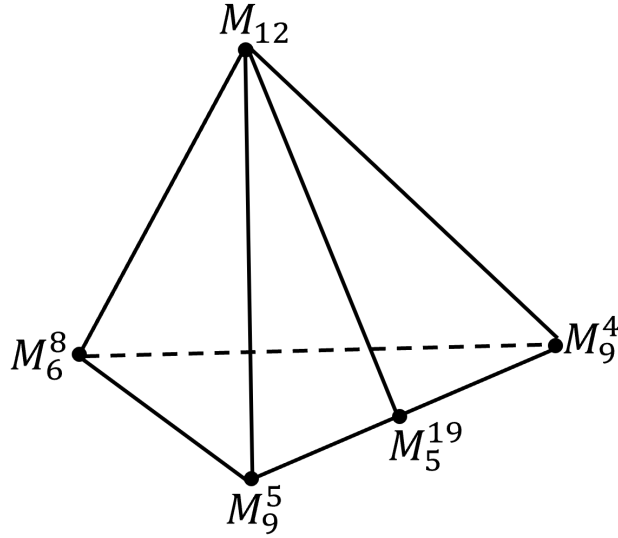


FIGURE 1.  $M_{12}M_9^5M_9^4M_5^{19}$  :  $\text{Sys}(\{c_1, c_4, c_{10}, c_{11}, c_{12}\})$ (5-ring),  
 $M_{12}M_9^3M_9^4M_6^8$  :  $\text{Sys}(\{c_1, c_4, c_7, c_8, c_{10}, c_{11}\})$ (6-ring),  
 $M_{12}M_6^8p_3M_9^5$  :  $\text{Sys}(\{c_1, c_4, c_8, c_{10}, c_{11}\})$ (5-chain)

### 3.2. $\text{Sys}(\{c_1, c_4, c_{10}, c_{12}\})$ .

$$\mathcal{V}(\{c_1, c_4, c_{10}, c_{12}\}, M_{12}) =$$

$$\{k(-1, x, y, x - y - 1, 3 - x + y, x - y - 1)A^{-1} : k > 0, -1 \leq x \leq 3, y \geq -1, 2x - 3 \leq y \leq x\}$$

It has three faces:

$$\begin{aligned} &\{k(-1, x, 2x - 3, 2 - x, x, 2 - x)A^{-1} : k > 0, -1 \leq x \leq 3\} \\ &\{k(-1, x, x, -1, 3, -1)A^{-1} : k > 0, -1 \leq x \leq 3\} \\ &\{k(-1, x, -1, x, 2 - x, x)A^{-1} : k > 0, -1 \leq x \leq 3\} \end{aligned}$$

which should look like Figure 2.

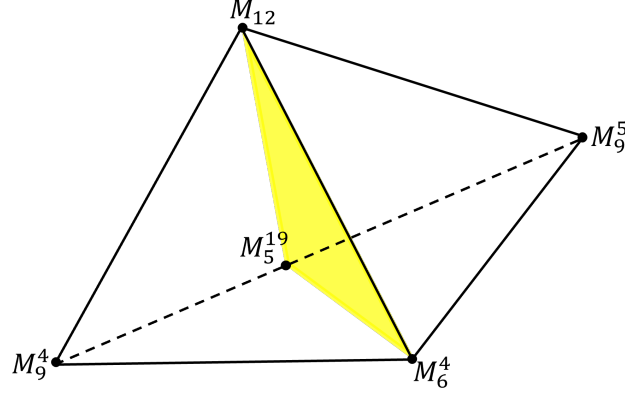


FIGURE 2.  $M_{12}M_9^5M_9^4M_5^{19} : \text{Sys}(\{c_1, c_4, c_{10}, c_{11}, c_{12}\})(5\text{-ring})$ ,  
 $M_{12}M_9^4M_6^4 : \text{Sys}(\{c_1, c_4, c_7, c_{10}, c_{12}\})(5\text{-chain})$ ,  
 $M_{12}M_9^5M_6^4 : \text{Sys}(\{c_1, c_4, c_6, c_{10}, c_{12}\})(5\text{-chain})$ .

### 3.3. $\text{Sys}(\{c_1, c_{10}, c_{11}, c_{12}\})$ .

$$\mathcal{V}(\{c_1, c_{10}, c_{11}, c_{12}\}, M_{12}) = \emptyset$$

$$\begin{aligned} \overline{\mathcal{V}(\{c_1, c_{10}, c_{11}, c_{12}\}, M_{12})} &= \overline{\mathcal{V}(\{c_1, c_4, c_5, c_8, c_{10}, c_{11}, c_{12}\}, M_{12})} \\ &= \{k(-1, x, x, -1, 3, -1) : k \geq 0, -1 \leq x \leq 3\} \end{aligned}$$

It's a 2-dimensional cone. So we need to use second order derivative to get more information about this cell.

To compute  $\mathcal{V}_2(\{c_1, c_{10}, c_{11}, c_{12}\}, \{c_1, c_4, c_5, c_8, c_{10}, c_{11}, c_{12}\}, M_{12})$ , let  $v_1 = k(-1, x, x, -1, 3, -1)A^{-1}$ , where  $k > 0$ ,  $-1 \leq x \leq 3$ . Suppose  $v_2 = (\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5, \lambda_6)$ , let  $f_i(x, k) = v_1 \cdot H_i(0) \cdot (v_1)^T$ , to compute  $v_2$ , we need to solve the following linear system:

$$\begin{aligned} v_2 \cdot dL(c_i)|_p + f_i(x, k) &= a, \text{ for } i = 1, 10, 11, 12, \\ v_2 \cdot dL(c_i)|_p + f_i(x, k) &> a, \text{ for } i = 4, 5, 8, \end{aligned}$$

where  $a \in \mathbb{R}$ .

By calculation, we can obtain:

$$\lambda_1 = \frac{1}{2}(f_{10} + f_{11} + f_{12} - 3a), \quad \lambda_3 = \frac{1}{2}(a - f_1), \quad a \in \mathbb{R},$$

$$\lambda_2 = (\sqrt{2} - 1)a + 2\sqrt{2 + 2\sqrt{2}}\lambda_4 + f_1 - \sqrt{2}f_{10},$$

$$\lambda_6 = \frac{1}{2\sqrt{2 + 2\sqrt{2}}} \left( -\sqrt{2}a - f_1 - f_{10} + f_{11} + (\sqrt{2} + 1)f_{12} - 2\sqrt{2 + 2\sqrt{2}}\lambda_5 \right),$$

$$(5) \quad \lambda_5 \geq \frac{1}{2\sqrt{2 + 2\sqrt{2}}} \left( (\sqrt{2} + 2)a + f_1 - f_{10} - f_{11} - f_{12} - \sqrt{2}f_4 \right),$$

$$(6) \quad \lambda_5 \leq \frac{1}{2\sqrt{2 + 2\sqrt{2}}} \left( (\sqrt{2} + 2)a - (\sqrt{2} + 1)f_{10} - (\sqrt{2} + 1)f_{11} - f_{12} + f_1 + \sqrt{2}f_8 \right),$$

$$\lambda_4 \in \mathbb{R}.$$

When equations 5 and 6 hold respectively, the sets consists of  $v_1$  and  $v_2$  can determine two faces respectively, then this stratum should look like Figure 3.

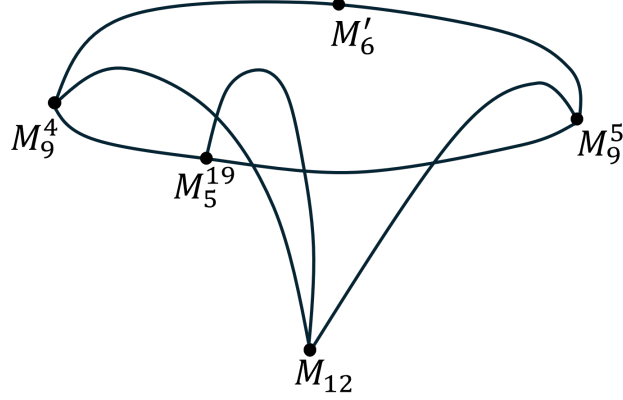


FIGURE 3.  $M_{12}M_9^5M_9^4M_5^{19} : \text{Sys}(\{c_1, c_4, c_{10}, c_{11}, c_{12}\})(5\text{-ring})$ ,  
 $M_{12}M_9^4M_9^5 : \text{Sys}(\{c_1, c_8, c_{10}, c_{11}, c_{12}\})(5\text{-chain})$ .

#### REFERENCES

- [1] N. An, J. Chen, I. Irmer, and P. Schaller. Thurston spine in genus 2.
- [2] N. An, F. Ihringer, and I. Irmer. Small genus, small index critical points of the systole function. *Experimental Mathematics*, 0(0):1–18, 2026.